

Translated

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Kuhn's Algorithm for Maximum Bipartite Matching

Problem

You are given a bipartite graph G containing n vertices and m edges. Find the maximum matching, i.e., select as many edges as possible so that no selected edge shares a vertex with any other selected edge.

Algorithm Description

Required Definitions

- A **matching** M is a set of pairwise non-adjacent edges of a graph (in other words, no more than one edge from the set should be incident to any vertex of the graph M). The **cardinality** of a matching is the number of edges in it. All those vertices that have an adjacent edge from the matching (i.e., which have degree exactly one in the subgraph formed by M) are called **saturated** by this matching.
- A **maximal matching** is a matching M of a graph G that is not a subset of any other matching.
- A **maximum matching** (also known as maximum-cardinality matching) is a matching that contains the largest possible number of edges. Every maximum matching is a maximal matching.
- A **path** of length k here means a *simple* path (i.e. not containing repeated vertices or edges) containing k edges, unless specified otherwise.
- An **alternating path** (in a bipartite graph, with respect to some matching) is a path in which the edges alternately belong / do not belong to the matching.
- An **augmenting path** (in a bipartite graph, with respect to some matching) is an alternating path whose initial and final vertices are unsaturated, i.e., they do not belong in the matching.